

Peter Asanov

SENIOR SOFTWARE ENGINEER

Amsterdam, Netherlands

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About

High-impact software engineering professional with 20+ years of experience, including over a decade building distributed systems for real-time, data-intensive, and mission-critical applications. Navigates complex technical problems from architecture to implementation, delivering clear, reliable, and elegant solutions.

Education

B.Sc. in Computer Science and Engineering

NOVOSIBIRSK STATE TECHNICAL UNIVERSITY (NSTU)

Novosibirsk, Russia

2002 – 2007

Skills

Distributed Systems	systems design, scalability, HA, fault tolerance, consensus protocols (Paxos, Raft), CAP trade-offs, event-driven systems
Dev stacks	
C++11-23	boost, flatbuffers, Cap'n Proto, asio, gtest, seastar
Go	gin, zap/slog, gomock
Python	FastAPI, mangum, uvicorn, SQLAlchemy, celery, dramatiq, pytest/pytest-mock, numpy, pandas, polars, PySpark, seaborn, pyo3/maturin
JS/TypeScript	React, Auth.js, tRPC, Prisma, Vite, Tailwind CSS, Axios
Rust	tokio, Hyper, Reqwest, clap
JVM	Java, Kotlin, Spring / Boot, Akka, Hibernate, NIO, Quarkus
GenAI / LLM	
RAG systems	retrieval-ready data design, hierarchical + reference schema modeling, context preparation for LLM calls, reproducible data transformation pipelines
LLM integration	protocol interoperability for agentic systems, OAuth2/OIDC for AI partner integrations, signed payload validation in model passthrough, debugging proprietary API integrations
Data quality for AI	circular dependency resolution, orphaned reference reconciliation, ambiguity handling, schema drift resilience
Build Systems	Nix, CMake, Ninja, Maven, Meson, Bazel
Cloud Native	
IAM/IdPs	AAD(enterprise), RBAC (Azure), KMS (AWS & GCP), Cognito, Auth0
Kubernetes	k3s, AKS, EKS, GKE, OpenShift
Other compute	Azure VM, ECS, EC2, GCE
Serverless	Azure Functions, AWS Lambda, GCF, Cloudflare Workers
IaC	Terraform, Helm, NixOps 4, Bicep, Azure Templates, CloudFormation, cdk
CI/CD & GitOps	Flux CD, GitHub Actions, GitLab CI/CD, ArgoCD, Azure DevOps
Observability	Azure Monitor / App Insights, AWS CloudWatch / X-Ray / CloudTrail, Prometheus / Alertmanager, Grafana / LGTM, OTLP
DevSecOps	Continuous Security, OWASP Top-10, Secure-by-Design, Zero Trust, PoLP, RBAC, Tenant Isolation, Intrusion Detection
Data Systems	

DBMS	PostgreSQL, PostGIS, TimescaleDB, MySQL, Cassandra, MongoDB, QuestDB, CosmosDB, RDS, DynamoDB, GCS, CloudSQL
KV-store / caching	etcd, Redis, RocksDB, Valkey, Memcached
Block storage	S3 / MinIO / RustFS, ABS
Queueing	RabbitMQ, ActiveMQ, Kafka
Big Data	Apache Spark, FLiP (Flink + NiFi + Pulsar), Kafka (Streaming), Airflow, DuckDB
Crypto	PKI, Hashes, Symmetric (block) ciphers, KDFs, ECC / Curve25519, ECDH, EdDSA, Ristretto

Work Experience

Senior Software Engineer

Rotterdam, Netherlands

THE OCEAN CLEANUP

March 2026 - July 2026

1. Re-engineered and owned the infrastructure for Onboard Compute platform - edge distributed system, managing UAV and vessel data ingestion, operational monitoring, and offshore cleanup planning.

Replaced the single-node k3s deployment with a three-node HA cluster and configured Kubernetes API failover between its two control-plane servers. Introduced GitOps for in-cluster state and built the host-provisioning and k3s/Flux bootstrap layer in Ansible. The GitOps workflow managed SOPS-encrypted runtime secrets and validated rendered manifests. Replicated this setup in a three-node local development environment. Automated node-access provisioning, wrote setup and recovery runbooks, and supported engineers through handover.

2. Alongside the platform work, helped clear the frontend backlog by refining proposed map fixes to improve legibility: higher-contrast density bands and a clearer hierarchy between vessel and route layers.

Platform: k3s, Flux CD, Kube-VIP, MetalLB (L2), Tailscale, Traefik, CloudNativePG, PostgreSQL/PostGIS/TimescaleDB, Redis Streams/Zenoh, Prometheus/Grafana, RustFS

Tools: Python, Ansible, Kustomize, SOPS, Terraform/Talos (PoC), Docker Engine, Tart, Rosetta, binfmt_misc, Nix/flake-parts (modular reproducible dev environments), Vue/TypeScript

Achievements:

- built the local cluster on Apple Silicon to run amd64 workloads via Docker Engine and Rosetta registered with binfmt_misc after debugging K3s's embedded-containerd path at the OCI/seccomp layer
- expanded Flux reconciliation, ingress, and remote-write alerts; instrumented UAV ingestion for data freshness, stream processing, dead-letter handling, and latency
- delivered HA ingress through a single VIP reachable over both the onboard LAN and tailnet
- replaced ingress-nginx with Traefik and authored the in-review routing RFC proposing Gateway API as the long-term model
- prototyped API-driven edge-cluster setup without node SSH using Terraform/Talos; the team retained k3s/Ansible

Senior Software Engineer | Partner Integrations

Amsterdam, Netherlands

Vox AI

July 2025 - Oct 2025

Built backend integrations for a voice-AI drive-thru platform, working directly with founders under two-week delivery windows.

1. Streamorders.com DSP integration: OAuth 2.0 server and location-management API

Implemented AWS-backed serverless infrastructure for the partner data-exchange protocol, with security, observability, and fast deployment cycles built in.

AWS Cognito-based OAuth 2.0/OIDC service with PKCE enforcement and compatibility handling for non-compliant partner clients

- correlation ID system for end-to-end tracing across authorize/token/webhook flows
- multi-tenant location management API with DynamoDB repository pattern and AWS X-Ray integration
- stable CDK stack layout to avoid 15-40 min Cognito/CloudFront reprovisioning during iteration
- comprehensive observability: X-Ray distributed tracing, correlation IDs in logs/headers, Lambda Powertools, EventBridge + SQS for OAuth event monitoring

2. POS menu transformation system for Redcat integration

Built a deterministic pipeline that converted flat POS data into nested and reference structures for voice-ordering RAG. Resolved ambiguity, circular dependencies, and orphaned references with transformations resilient to upstream schema changes.

3. RAG data quality engineering + CI integration

Converted noisy POS inputs into retrieval-ready structures and added CI validation to catch regressions from schema and data changes.

Platform: AWS Lambda, API Gateway, Cognito, DynamoDB, X-Ray, CloudWatch, EventBridge, SQS, CDK (IaC)

Tools: TypeScript, Python 3.12+, Pydantic, FastAPI, mangum, moto, pytest/pytest-asyncio, vitest, httpx, uv

Achievements:

- delivered working OAuth 2.0 integration bridging AWS Cognito with third-party DSP protocol under 2-week timeline
- built production-ready, RAG-oriented menu mapper handling poor data quality and complex hierarchical/reference relationships with strong test coverage
- implemented correlation ID tracing enabling sub-second visibility across distributed OAuth flows
- integrated RAG data quality checks into CI workflows, reducing regressions from upstream POS schema and data changes

Senior Software Engineer

Netherlands

FREELANCE / CONTRACTING

2022 – Present

Delivered platform, data, and systems engineering across client and open-source projects:

- Open-source contribution (tailscale/tailscale): landed Unix socket proxy support for `tailscale serve unix:/path/to/socket` in `cmd/tailscale + ipn`; enables exposing services like Docker/containerd/PHP-FPM over Tailscale without TCP bridging, with safeguards for sensitive local sockets
- Open-source contribution (blacktop/ipswn): fixed demangling of internal symbols (including `.cold.*`) to improve readability and correctness of disassembly output
- Agentic protocol contribution (opencode-antigravity-auth): reverse-engineered signed payload handling for Claude 4.6 interleaved thinking in private Google Antigravity API passthrough; preserved valid signatures and stripped only invalid ones to prevent model instability and reasoning leakage
- Deliverect, full-cycle restaurant automation PaaS: extended ordering data model and API behavior
Tools: Python, Eve, Celery, Kubernetes (GKE), TypeScript, React, Redis, MinIO (local backend), Terraform, Helm
- Ahold Delhaize (Albert Heijn): online data-ingestion platform for internal compliance and R&D analytics
Tools: Go, Kubernetes (AKS, Kafka Strimzi Operator), GitOps (Argo CD), Airflow, Spark, Terraform, Helm
- Private customer: taxi-terminal telemetry ingestion for compliance and fraud analysis
Tools: Go, Kubernetes (AKS), Kafka, PostgreSQL, Terraform, Helm
- LEGO: internal IAM / identity-provider functionality
Tools: Azure Functions, Azure SAML identity provider, Python
- Private customer: cloud-based video-conversion service
Tools: C++20, CMake, Ninja, Go, gRPC-Gateway, FFmpeg, Kubernetes (EKS), CloudWatch, Airflow, Nix, Terraform (Terranix), Helm
- Private customer: custom video calls service
Tools: C++23, CMake, Ninja, Asio, libdatachannel (WebRTC stack), Nix, Ansible
- Pet project [in development]: configurable HFT matching engine focused on high-throughput, low-latency order matching for exchanges and arbitrage systems
Tools: C++23, SIMD intrinsics, Cap'n Proto, Go

Core Software Engineer

Haarlem, Netherlands

AIMMS B.V.

Nov 2019 – Sep 2021

- led core C++ platform work in an AWS-to-Azure migration, replacing Java/Scala services with a modern C++ service framework
- built a C++17 microservice template with an API-first protobuf DSL used for both gRPC and REST surfaces
- bootstrapped Azure landing-zone foundations for Terraform state and RBAC
- built migration tooling for sensitive customer data
- wrote a declarative-schema CRUD generator for administrative and internal tools, reducing repetitive R&D work

Tools: ECS, EC2, CloudWatch, Go, Python, Java, Scala, C++17, TypeScript, React, Conan, CMake, Ninja, gRPC-Gateway, Kubernetes (AKS), Azure Application Insights, Terraform, Helm

Achievements: tackled cost-efficiency and throughput predictability of compute-heavy optimization workloads (with non-deterministic convergence) by implementing Kubernetes-native solutions: bin packing via resource requests/limits and affinity rules, priority classes, and a pre-warmed spare node strategy using Cluster Autoscaler and taints/tolerations

Senior Software Engineer

Amsterdam, Netherlands

HVR SOFTWARE (ACQUIRED BY FIVETRAN)

Jan 2018 – Oct 2019

Built features and fixes for HVR's high-throughput, log-based data-replication engine.

- contributed to product core replication workflows
- reverse-engineered Oracle histogram metadata to support custom sharding functionality
- built a Go REST service for cloud integration

Tools: C99, Python, Go, multiple RDBMS and Big Data stacks (integration), Oracle (histogram internals), Airflow, Spark

Achievements: rewrote a core feature-flag transpiler used in HVR's CI/CD pipeline, reducing runtime from over 1 minute to 5 seconds on a critical build path

Software Engineer

Eindhoven, Netherlands

MAPSCAPE B.V.

2017

Implemented backend components of the Mapscape NDS compiler for raw map-data parsing and added parallel parsing to reduce processing time.

Tools: C++11/14, Boost, GTest, SQLite, shell, Python, Airflow

Senior Software Engineer

St. Petersburg, Russia

NEW CLOUD TECHNOLOGIES

2014 – 2017

Built collaboration-engine features for an on-prem office suite and developed the spreadsheet client for Samsung Tizen.

Tools: C++14, CMake, Emscripten, Objective-C, Qt, SWIG, Kubernetes

Achievements:

- built a cross-platform FreeType-based font-rendering engine with subpixel positioning for scalable document layouts and integrated it into a large enterprise codebase
- developed advanced spreadsheet editing for Samsung Tizen using a declarative, compile-time state machine to avoid dynamic dispatch overhead and resolve editing edge cases observed in competing web suites

Senior Software Engineer

St. Petersburg, Russia

AIS

2014

Built C++11 microservices for a crypto-mining-pool banking and exchange subsystem:

- designed an Asio-based service framework
- implemented multi-stage, fault-tolerant transactions
- reduced direct-loss risk with append-only storage, two-factor transactions, circuit breakers, alerting, rollback paths, and extensive testing
- shipped the zero-to-one production release with a three-engineer team, owning the C++ subsystem end to end

Tools: C++11, Core Java, Groovy, Ruby (Chef)

Achievements:

- designed and implemented all C++11 services for the crypto exchange banking facility, including multi-stage transactions and circuit breakers with rollback paths

- delivered the production launch in under two months with personal accountability for reliability under financial risk

Previous work history

AVAILABLE UPON REQUEST

2005 – 2014